

■ BusinessNext

AWS Cost Estimator

Pricing & Infrastructure Report

HDFC Bank

Mode: On-Premise — Client Hosted

Date: 18 March 2026

Inflation: 4% / yr

AWS Region: us-east-1

Executive Summary

High-level cost overview and infrastructure snapshot

9	96	172	224
Worker Nodes	vCPUs	RAM (GB)	DB RAM (GB)
Total compute nodes	Total virtual CPUs	Total memory	Database memory

Project Details

Customer / Bank	HDFC Bank
Client Mode	On-Premise — Client Hosted
Database Type	PostgreSQL
Total Named Users	15,500
Concurrent Users	4,650 (30% of named)
Mobile Users	4,050
Worker Nodes	9
Total vCPUs (worker)	96
Total RAM GB (worker)	172 GB
Data Size	2100 GB
Estimate Date	18 March 2026

Node Distribution

AI-assisted infrastructure architecture

Total Worker Nodes	9
Total DB Nodes	11
Fixed / Managed Roles	7
Confidence	Rule-Based (All Groq Models Failed)
Distribution Method	Groq LLM + Rule-based

Category	Role / Service	Nodes	Instance Fam.	vCPU/node	RAM GB/node	Model
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	5	Compute Intensive	16	32	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for Graphana & Prometheus	2	Compute Intensive	8	16	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK - optional	2	Compute Intensive	8	16	Reserved 3 Yr
PGSQL Database	Linux Nodes – Database server (Primary)	2	Memory Intensive	32	128	None
PGSQL Database	Linux Nodes – Cluster (etcd+haproxy, pgbackrest)	4	Memory Intensive	2	8	None
PGSQL Database	Storage: SAN (Primary DB)	2	10K IOPS	—	—	None
DMS	DMS Storage + Replication	1	Cloud Storage Service	—	—	On Demand
Security & Compliance	WAF + AWS Shield Advanced	1	Managed Service	—	—	On Demand
Security & Compliance	CloudTrail + Config Audit Logs	1	Managed Service	—	—	On Demand
Kubernetes Cluster	Cloud Managed K8s	1	K8s Service	—	—	None
Caching & Session	Elastichache Service	2	Memory Optimized	4	16	Reserved 3 Yr
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	—	—	On Demand
Infrastructure & Monitoring	Bastion Host	1	General Purpose	2	4	Reserved 3 Yr
Infrastructure & Monitoring	NAT Gateway	2	NAT	—	—	On Demand
Infrastructure & Monitoring	Image Registry (ECR)	1	Managed Service	—	—	On Demand
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage Service	—	—	On Demand

Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

Disaster Recovery (DR) Environment

Monthly	Annual	5-Year
\$9,504	\$114,052	\$642,454

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s (DR)	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI) – DR	4	m5.4xlarge	\$2,324
Kubernetes	Worker Nodes (Grafana) – DR	2	m5.2xlarge	\$643
Kubernetes	Worker Nodes (EFK) – DR	2	m5.2xlarge	\$724
PGSQL DB	DB Server Primary – DR	2	m5.8xlarge	\$2,291
PGSQL DB	DB Cluster (etcd+haproxy) – DR	4	m5.large	\$312
PGSQL DB	Storage: SAN Primary – DR	2	EBS io2	\$750
PGSQL DB	Reporting DB – DR	1	m5.8xlarge	\$1,145
PGSQL DB	Storage: SAN Reporting – DR	1	EBS io2	\$375
DMS	DMS Storage + Replication – DR	1	dms.c5.large	\$296
Caching	Elasticache Service – DR	2	cache.r6g.large	\$242
Infrastructure	Load Balancer – DR	2	ALB	\$33
Infrastructure	NAT Gateway – DR	2	NAT	\$32
Infrastructure	Bastion Host – DR	1	m5.large	\$94
Infrastructure	Image Registry (ECR) – DR	1	ECR	\$51
Backup	Back Up – DR	1	S3	\$118

Notes & Assumptions

AWS Pricing	All prices are AWS On-Demand, us-east-1 region, Linux/UNIX, shared tenancy. Reserved or Savings Plans pricing will reduce costs by 30–40%.
Inflation Rate	4% per year applied to all compute costs from Year 2 onwards.
Fallback Rates	If live AWS Pricing API is unavailable, hardcoded March 2026 estimates are used.
Node Distribution	Distribution computed via rule-based algorithm enhanced by Groq LLM (llama-3.3-70b-versatile) at temperature=0. SHA-256 cache prevents redundant calls.
Security Costs	Airtel SOC (\$500/mo), SOC Machines (Rs 30,000/yr), Req 4 Antivirus (Rs 2,600/yr), Req 6 Data Discovery (Rs 4,000/yr) are fixed INR-converted estimates.
Pre-Prod / DR	Pricing derived from reference sizing workbook. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server and Oracle use AWS RDS Managed.
Named Users	Year-over-year growth assumed at 5%. Concurrent users assumed at 30% of named users.
GCP Pricing	GCP prices are estimates using hardcoded us-central1 On-Demand rates (March 2026). GCP Cloud Billing API integration available via GOOGLE_CLOUD_PROJECT env var.
Confidentiality	This document contains confidential cost estimates. Do not distribute outside the project team without authorisation.

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